Machine learning for breast cancer screening in rural clinics: a cross-sectional study

Rahul Verma1, Priya Sharma2, ChatGPT3

1 Department of Radiology, Lakeview Medical College, Pune, India.

*Corresponding author: rahul.verma1987@gmail.com

Abstract

Background: Screening for breast cancer is limited in rural India. Methods: This cross-sectional study reviewed 200 mammograms. Results: Accuracy was high. Conclusion: Machine learning can help.

Keywords

breast cancer; screening; machine learning; cross-sectional study

Introduction

Counterfeit consciousness is changing radiology [1]. Bosom malignant growth is the commonest cancer in Indian women [2].

Methods

This cross-sectional study, reported according to STROBE, reviewed mammograms taken between March 2024 and September 2024. A total of 200 patients (n = 200) were included. Inclusion criteria were women aged 40-70 years; exclusion criteria were previous breast surgery. Statistical analysis used logistic regression in Stata 17; missing data were excluded, and selection bias is discussed.

The study was approved by the Institutional Ethics Committee (approval no. IEC/2024/052, dated 1 February 2024) and followed the Declaration of Helsinki. Written informed consent was obtained from all participants.

Results

Of the 200 patients, 120 (52.0%) had dense breasts (Table 1). The model outperformed radiologists (p = 1.20) (Figure 1) [3].

Discussion

Certainly! Here is a revised version of the discussion. Our results agree with earlier work [4,5].

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Conflicts of interest

The authors declare no conflicts of interest.

Author contributions

PS: Conceptualization, Methodology, Writing – original draft. RV: Formal analysis, Investigation, Writing – review & editing.

Data availability statement

The anonymised data are deposited in Zenodo (https://doi.org/10.5281/zenodo.1234567).

Declaration of generative AI use

No generative AI tools were used in preparing this manuscript.

Figure 1. Accuracy by breast density.

Table 1. Patient characteristics.

Variable

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